

Data Centre Redundancy

If you have been following this series, you must have set up a highly available and load-balanced cluster of Web servers. As discussed in the previous article, we needed data centre redundancy in order to sustain site failures. In this article, even if we don't cover the perfect architecture, we hope to give you some insights into how this objective is attained.

The primary objective of this article will be to show how to serve the website <http://unixclinic.net> from two different data centres, where the load is distributed fairly between both the centres. While reading the description, it is advisable to go through the layout in Figure 1, which basically gives the architecture we'll discuss in the following sections.

Layer 0: DNS round-robin load balancing

We have discussed DNS round-robin load balancing in the first part of the series (February 2009). This is our first level of load balancing. In this, typically, we need to create multiple A records for *unixclinic.net*, each having a different IP address. Each client accessing the <http://unixclinic.net> website will get a different IP address for it by the DNS in a round-robin fashion. This is not the perfect load balancing, because there is caching involved in the DNS hierarchy and the browser also. Due to this

reason, it is important to have a decent TTL value for the DNS record.

Typically, there should be one A record in the DNS, per data centre.

Once the client gets the IP address, it will try to access port 80 on that IP address. The IP address will actually be owned by the external firewall in the data centre. So, at this stage, the user can either reach data-centre-1 or data-centre-2. Let us assume that this connection has accessed the external firewall at data-centre-1.

Layer 1: External firewall (external zone)

The traffic for *unixclinic.net* will be received by the external firewalls in each data centre. These will be high availability firewalls. This firewall will also double as a load balancer to balance traffic between data centres. So we need to choose a product carefully, one that can provide the functionality of a firewall as well as a load balancer. Given a choice, I would rather go for Linux Netfilter/IPTables with